

Chapter 37: Degree Centrality in Network Analysis: Theory, Variants and Applications

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Abstract

Degree centrality is a fundamental network measure that evaluates a vertex through its number of direct connections. This chapter presents its graph-theoretic formulation for undirected and directed networks, together with normalised degree and Freeman's network-level degree centralisation. Important extensions, including weighted, signed, bipartite, temporal, multilayer, and group degree measures, are briefly examined. The principal advantages and limitations of degree centrality are discussed, particularly its simplicity, interpretability, local character, and sensitivity to network construction. Selected applications in social, biological, epidemiological, transportation, citation, and collaboration networks are also considered. Degree centrality remains an effective baseline for identifying locally prominent vertices when direct interaction is the main feature of interest.

Keywords: degree centrality; network analysis; graph theory; node importance; complex networks; network centralisation

1. Introduction

Networks provide a mathematical framework for representing systems composed of interacting entities. The entities are modelled as vertices, while their relationships are represented by edges. This approach is widely used in the study of social interactions, transportation systems, biological processes, communication networks, scientific collaboration, and digital information structures (Newman, 2010).

A central task in network analysis is to identify vertices that occupy important structural positions. Different centrality measures capture different forms of importance, such as direct connectivity, proximity, brokerage, or association with influential neighbours (Freeman, 1979; Newman, 2010). Degree centrality is the simplest of these measures because it evaluates a vertex through the number of edges incident on it.

The measure is especially useful when direct interaction is the main process of interest. A vertex with many neighbours has greater immediate opportunity for communication, exchange, exposure, or collaboration. However, degree centrality does not account for indirect paths, global accessibility, or the structural importance of neighbouring vertices. Its interpretation must therefore remain consistent with the type of network and the process being studied (Borgatti, 2005; Rodrigues, 2019).

This chapter presents the mathematical foundations, principal variants, computational aspects, applications, and recent developments of degree centrality. It also examines its main advantages and limitations as a measure of local network prominence.

2. Fundamentals of Network Theory

Network theory provides the mathematical foundation for representing and analysing systems composed of interacting entities. Such systems are naturally modelled using graphs, where the entities are represented by vertices (or nodes) and their interactions are represented by edges (or links). This graph-theoretic representation enables the application of mathematical methods to investigate structural properties, connectivity patterns, and the relative importance of vertices within a network (Newman, 2010).

Formally, a graph is defined as

$$G = (V, E),$$

where $V = \{v_1, v_2, v_3, \dots, v_n\}$ denotes the finite set of vertices (or nodes), and $E \subseteq V \times V$ represents the set of edges (or links) connecting pairs of vertices. The number of vertices $n = |V|$ is called the order of the graph, while the number of edges $m = |E|$ is called its size. Graphs may be undirected, when edges have no orientation, or directed, when each edge has a specified direction. Networks may also be unweighted or weighted, depending on whether edges record only the presence of a connection or carry a numerical value such as frequency, strength, capacity, or distance (Newman, 2010).

A convenient mathematical representation of a network is provided by its adjacency matrix,

$$A = (a_{ij})_{n \times n}$$

where

$$a_{ij} = \begin{cases} 1, & \text{if an edge exists between } v_i \text{ and } v_j, \\ 0, & \text{otherwise} \end{cases}$$

For a simple graph, $a_{ii} = 0$. In an undirected graph,

$$A = A^T,$$

whereas the adjacency matrix of a directed graph need not be symmetric.

For weighted networks, the binary entries are replaced by the corresponding edge weights, producing a weight matrix $W = (w_{ij})$.

The adjacency matrix forms the basis for many computational algorithms and provides an efficient representation for analysing network structure (Newman, 2010).

One of the most fundamental structural properties of a graph is the degree of a vertex. In an undirected graph, the degree of vertex the degree of a vertex v_i , denoted by d_i , or $\deg v_i$ is the number of edges incident to that vertex:

$$d_i = \sum_{j=1}^n a_{ij}.$$

The degrees of all vertices satisfy the handshaking lemma,

$$\sum_{i=1}^n d_i = 2m,$$

because every undirected edge contributes one unit to the degree of each endpoint. Hence, the average degree is

$$\bar{d} = \frac{1}{n} \sum_{i=1}^n d_i = \frac{2m}{n}.$$

For directed graphs, two degree measures are defined: the in-degree, representing the number of incoming edges, and the out-degree, representing the number of outgoing edges. These concepts provide the mathematical basis for Degree Centrality, which quantifies the local importance of a vertex through its immediate connectivity (Freeman, 1979; Newman, 2010).

3. Degree Centrality: Concept and Mathematical Formulation

Degree centrality measures the local prominence of a vertex through its number of direct connections.

For an undirected graph $G = (V, E)$ with n vertices, the degree of a vertex v_i is

$$d_i = \sum_{j=1}^n a_{ij},$$

where $A = (a_{ij})$ is the adjacency matrix defined in section 2.

The Degree Centrality of vertex v_i is therefore

$$C_D(v_i) = d_i.$$

For a simple undirected graph,

$$0 \leq C_D(v_i) \leq n - 1.$$

A vertex with degree 0 is isolated, whereas a vertex with degree $n - 1$ is adjacent to every other vertex in the graph.

To facilitate comparisons among networks of different sizes, Freeman (1979) proposed the normalised Degree Centrality,

$$C_D^*(v_i) = \frac{d_i}{n - 1},$$

where $n - 1$ represents the maximum possible degree of a vertex in a simple graph. Consequently,

$$0 \leq C_D^*(v_i) \leq 1.$$

A value close to one indicates that the vertex is directly connected to most other vertices in the network, whereas values near zero correspond to poorly connected or peripheral vertices (Freeman, 1979).

3.1 Degree Centrality in Directed Networks

For directed networks, Degree Centrality is divided into in-degree and out-degree centralities. The in-degree of a vertex measures the number of incoming edges,

$$d_i^{in} = \sum_{j=1}^n a_{ji},$$

while the out-degree measures the number of outgoing edges,

$$d_i^{out} = \sum_{j=1}^n a_{ij}.$$

Accordingly, the in-degree and out-degree centralities are

$$C_D^{in}(v_i) = d_i^{in},$$

and

$$C_D^{out}(v_i) = d_i^{out}$$

Their normalised forms are

$$C_D^{in*}(v_i) = \frac{d_i^{in}}{n - 1},$$

and

$$C_D^{out*}(v_i) = \frac{d_i^{out}}{n - 1},$$

These measures often provide different insights. In social media networks, for example, in-degree may represent popularity through the number of followers, whereas out-degree reflects the number of individuals followed by a user. Similarly, in citation networks, in-degree corresponds to the number of citations received by a publication and is frequently regarded as an indicator of scholarly influence (Newman, 2010).

3.2 Network Degree Centralisation

Degree centrality refers to the prominence of an individual vertex. Freeman (1979) also introduced degree centralisation to describe the extent to which the connectivity of an entire network is concentrated around one or a few vertices.

Let $d_{max} = \max_{1 \leq i \leq n} d_i$.

For a simple undirected graph, network degree centralisation is defined as

$$C_D(G) = \frac{\sum_{i=1}^n (d_{max} - d_i)}{(n - 1)(n - 2)}$$

The denominator is the maximum possible sum of degree differences for a graph of order n , attained by the star graph. Therefore,

$$0 \leq C_D(G) \leq 1.$$

A value of 1 indicates a maximally centralised star network, while a value of 0 occurs when all vertices have the same degree, as in a regular graph.

To illustrate the computation of degree centrality, consider the undirected network shown in Figure 1.

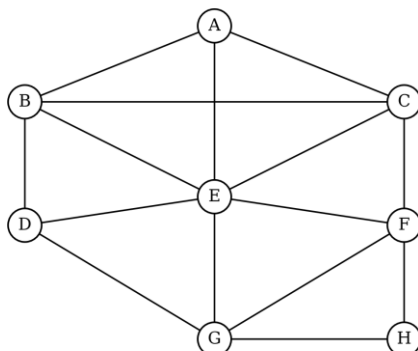


Figure 1. Illustrative undirected network used for computing degree centrality
The graph has eight vertices, and its degree sequence is (3,4,4,3,6,4,4,2). Since $n = 8$, the maximum possible degree is $n - 1 = 7$. The degree and normalised degree centralities are shown in Table 1.

Table 1. Degree of vertices in the illustrative network

Vertex	Degree (d_i)	Normalised Degree Centrality (C_D^*)
A	3	$3/7=0.429$
B	4	$4/7=0.571$
C	4	$4/7=0.571$
D	3	$3/7=0.429$
E	6	$6/7=0.857$
F	4	$4/7=0.571$
G	4	$4/7=0.571$
H	2	$2/7=0.286$

From Table 1, vertex E possesses the highest Degree Centrality whereas vertex H has the smallest Degree Centrality. These values describe immediate connectivity only and do not account for indirect paths or the wider network structure.

For this graph, $d_{max} = 6$.

Hence,

$$C_D(G) = \frac{(6-3) + (6-4) + (6-4) + (6-3) + (6-6) + (6-4) + (6-4) + (6-2)}{(8-1)(8-2)}$$

$$= \frac{18}{42} \approx 0.429$$

Thus, the illustrative network exhibits a moderate degree of centralisation.

4. Variants of Degree Centrality

The basic degree measure can be extended to networks in which connections possess weights or signs, vertices belong to different classes, interactions change over time, or several relational layers coexist. These variants retain the local character of degree centrality while adapting its interpretation to the structure of the network.

4.1 Weighted Degree Centrality

In a weighted network, each edge carries a numerical value representing interaction strength, frequency, capacity, or volume. Let $W = (w_{ij})$ be the corresponding weight matrix. The weighted degree, commonly called the strength of vertex v_i is

$$s_i = \sum_{j=1}^n w_{ij}.$$

Degree and strength measure different aspects of local connectivity: degree counts distinct neighbours, whereas strength gives the total weight of incident connections (Barrat et al., 2004). Opsahl et al. (2010) proposed a generalised degree measure,

$$C_D^{(w,\alpha)}(v_i) = d_i^{1-\alpha} s_i^\alpha,$$

where $\alpha \geq 0$ controls the relative influence of the number and weight of ties. When $\alpha = 0$, the measure reduces to ordinary degree, while $\alpha = 1$ gives node strength. This formulation assumes that larger weights indicate stronger or more substantial relations; weights representing distance or cost require different interpretation.

4.2 Signed Degree Centrality

A signed network contains both positive and negative relations, such as cooperation and conflict or trust and distrust. The positive and negative degrees of v_i may be defined as

$$d_i^+ = \sum_{j=1}^n 1(w_{ij} > 0), d_i^- = \sum_{j=1}^n 1(w_{ij} < 0)$$

where $1(\cdot)$ is the indicator function. These quantities should normally be reported separately because a single net value may conceal the simultaneous presence of many positive and negative ties. More advanced signed-network centrality measures additionally incorporate direction, structural balance, and indirect relations (Gromov, 2025).

4.3 Bipartite Degree Centrality

In a bipartite graph $G = (U, W, E)$, edges occur only between the two disjoint vertex classes U and W . Consequently, the maximum possible degree of a vertex depends on the size of the opposite class. For $u \in U$,

$$C_D^*(u) = \frac{d(u)}{|W|},$$

whereas for $w \in W$,

$$C_D^*(w) = \frac{d(w)}{|U|}.$$

This normalisation is appropriate for networks such as author-publication, customer-product, and student-course networks. It should be calculated on the original bipartite graph because one-mode projection may create additional derived connections and alter degree values.

4.4 Temporal and Multilayer Degree Centrality

In a temporal network, edges vary with time. If $A(t)$ denotes the adjacency matrix at time t , the instantaneous degree of v_i is

$$d_i(t) = \sum_{j=1}^n a_{ij}(t).$$

Degree may also be computed over a specified time interval or weighted according to the recency or duration of interactions. Such formulations retain temporal information that may be lost when all observations are combined into a single static network (Holme & Saramäki, 2012).

In a multilayer network, each layer represents a distinct relation, context, or period. If $d_i^{[\ell]}$ is the degree of v_i in layer ℓ , its layer-degree vector is

$$\mathbf{d}_i = (d_i^{[1]}, d_i^{[2]}, \dots, d_i^{[L]}).$$

An aggregated measure is the overlapping degree

$$o_i = \sum_{\ell=1}^L d_i^{[\ell]}.$$

The full vector may be more informative than the sum because vertices with identical overlapping degrees can have different distributions of connectivity across layers (Kivelä et al., 2014).

4.5 Group Degree Centrality

For a group of vertices $S \subseteq V$ group degree centrality counts the vertices outside the group that are adjacent to at least one group member:

$$C_D(S) = |(\bigcup_{v_i \in S} N(v_i)) \setminus S|.$$

Its normalised form is

$$C_D^*(S) = \frac{C_D(S)}{n - |S|}.$$

Group degree centrality measures direct external coverage and is useful for selecting teams, service centres, monitoring units, or intervention groups (Everett & Borgatti, 1999).

5. Advantages and Limitations

Degree centrality is simple, interpretable, and computationally efficient. It is useful for identifying highly connected vertices and can be applied to several network types. However, it measures only direct connectivity and does not capture indirect paths, brokerage, global accessibility, or the importance of neighbouring vertices. Its interpretation is also sensitive to network construction, missing data, and the meaning of edge weights (Borgatti, 2005; Freeman, 1979; Newman, 2010; Opsahl et al., 2010).

Table 2. Summary of the advantages and limitations of degree centrality

Aspect	Advantage	Limitation
Interpretation	Simple measure of direct connectivity	High degree does not always imply global importance
Information required	Uses only immediate-neighbour information	Ignores indirect connections and broader topology

Computation	Fast and suitable for large sparse networks	Complete ranking may still require network-wide comparison
Hub identification	Detects highly connected vertices	Cannot determine whether neighbours are themselves important
Flexibility	Extendable to directed, weighted, temporal, and multilayer networks	Interpretation depends on direction, weight, time, and layer definitions
Practical use	Effective when direct interaction dominates	Less suitable for brokerage, prestige, and global accessibility
Comparability	Normalisation permits comparison within similarly defined networks	Results remain sensitive to network boundaries and data quality

6. Applications of Degree Centrality

Degree centrality is useful when the number of direct connections of a vertex is relevant to the problem under study. Its interpretation depends on the type of network and the meaning of its edges. A high degree may indicate popularity, activity, exposure, accessibility, or biological connectivity (Borgatti, 2005; Newman, 2010).

In social and communication networks, degree centrality identifies individuals or organisations with many immediate contacts. In directed networks, in-degree may indicate received attention or popularity, whereas out-degree may reflect communication activity (Newman, 2010).

In biological networks, highly connected proteins, genes, or metabolites may function as hubs. Degree-based screening can help identify potentially important biological components, although such findings require experimental or clinical validation. In epidemiological networks, individuals with many contacts may have greater opportunities to acquire or transmit infection (Keeling & Eames, 2005; Newman, 2010).

In transportation and infrastructure networks, high-degree vertices represent airports, stations, intersections, or facilities with many direct connections. These vertices may serve as local hubs, but a lower-degree vertex may still be structurally important if it connects otherwise separated regions (Guimerà et al., 2005).

In citation and collaboration networks, in-degree can represent citations received, while degree in a co-authorship network represents the number of direct collaborators. Such measures indicate immediate recognition or collaboration, but they should not be treated as complete measures of scholarly quality or influence (Newman, 2001, 2010).

Overall, degree centrality is most appropriate when direct interaction is the main mechanism of interest. When the objective concerns brokerage, long-range accessibility, prestige, or network-wide influence, it should be interpreted alongside other centrality measures (Borgatti, 2005; Saxena & Iyengar, 2020).

7. Conclusion

Degree centrality remains one of the most useful and accessible measures in network analysis because it provides a direct indication of a vertex's immediate connectivity. By counting incident edges, it identifies locally prominent vertices and can be extended to directed, weighted, signed, bipartite, temporal, multilayer,

and group-based networks. Its mathematical simplicity and ease of interpretation make it suitable for applications in social, biological, epidemiological, transportation, citation, and collaboration networks.

At the same time, degree centrality offers only a local view of network structure. It does not consider indirect paths, brokerage positions, global accessibility, or the importance of neighbouring vertices. Its values may also be influenced by network boundaries, missing links, and the way edges or weights are defined. Therefore, a high degree should not automatically be interpreted as complete or network-wide influence. The measure is most appropriate when direct interaction is central to the research problem and is best understood in relation to the structure, context, and purpose of the network being analysed.

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